# Population Aging and the Household Energy Portfolio: Evidence from U.S. Microdata

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**Abstract**

Population aging can affect household energy demand through several margins: residential fuel use, transport fuel demand, and exposure to energy-cost shocks. This paper studies how aging is associated with the household energy portfolio in the United States. The analysis combines Residential Energy Consumption Survey (RECS) microdata, Consumer Expenditure Survey (CEX) Interview public-use microdata from 2000–2024, state-year weather indicators from the Energy Information Administration, EIA state-year fuel prices, and Census state age structure estimates. The empirical design is organized around three layers of evidence. DiNardo–Fortin–Lemieux reweighting measures how much changing age composition accounts for long-run portfolio shifts. Cohort pseudo-panels examine whether elderly-household differences survive controls for cohort structure. Heating-degree-day shock interactions provide the main mechanism test for energy vulnerability. A survival-adjusted shift-share IV is reported as a supplementary state-level check rather than as the primary source of identification. The evidence indicates that aging is not the main driver of aggregate residential electrification: from 2000–2004 to 2020–2024, the electricity share of residential energy expenditure rose by 5.1 percentage points, while age-composition reweighting explains a -0.3 percentage point contribution in the age-only specification and is not robust to geographic and housing-related composition controls. Aging is more important for two other margins. First, age composition reduces gasoline reliance; the age-composition effect on the gasoline share is -0.88 percentage points, equivalent to about 42 percent of the observed 2.09 percentage point change in the opposite direction. Second, older households are more vulnerable to heating shocks. A one-standard-deviation increase in state-year heating-degree-day shocks raises elderly households' log home-energy expenditure by an additional 0.95 percent and increases their high-energy-burden risk by 0.87 percentage points. This burden response is larger among low-income elderly households. Price-adjusted expenditure indexes show that the residential fuel-share result is not fully explained by relative fuel-price movements, but they do not replace metered energy quantities. Overall, population aging has no robust explanatory role for residential electrification, but it is a more consequential source of household energy-portfolio reallocation and heating-related vulnerability.

**Keywords:** population aging; household energy portfolio; energy burden; transport fuel demand; weather shocks; DFL decomposition; cohort pseudo-panel

**JEL Codes:** J11; Q41; Q48; D12; C21

# 1 Introduction

Population aging is changing the demographic structure of advanced economies. At the same time, household energy systems are being transformed by electrification, fuel switching, changing transport behavior, rising cooling demand, and climate-related weather risk. These transitions are often studied separately. Yet aging can plausibly reshape household energy demand through several channels. Older households may spend more time at home, have different thermal comfort needs, live in older housing units, face fixed-income constraints, and drive less. These mechanisms imply that aging may change not only the level of energy spending, but also the allocation of household energy expenditure across residential fuels, transport fuels, and high-burden risk.

This paper asks how population aging is associated with the household energy portfolio in the United States. The term portfolio is used deliberately. The analysis does not treat household energy structure as synonymous with residential electrification. Instead, it distinguishes three margins: the residential allocation between electricity and fossil home-energy fuels, transport fuel reliance measured by gasoline expenditure, and energy vulnerability measured by home-energy burden and responses to heating shocks. This distinction is important because a small effect on the aggregate electricity share can coexist with larger effects on gasoline demand or vulnerability to unusually cold years.

The empirical analysis combines four sources of data. First, RECS household microdata from 2015 and 2020 are used to document cross-sectional differences between elderly and non-elderly households in residential energy bundles. Second, a long CEX Interview public-use microdata panel from 2000 to 2024 is constructed to measure household expenditures on electricity, natural gas, fuel oil, and gasoline. Third, EIA state-year heating-degree-day and cooling-degree-day indicators are merged to test whether older households respond differently to weather-driven heating demand. Fourth, Census state age distributions are used to construct state-year elderly shares and a predetermined age-structure shift-share instrument.

The paper makes three contributions. First, it provides microdata evidence on population aging and the household energy portfolio over a period of substantial change in U.S. residential and transport energy use. Second, it separates three empirically distinct margins—residential fuel mix, gasoline reliance, and energy-burden vulnerability—that are often compressed into a single measure of household energy demand. Third, it uses repeated cross-sections, cohort pseudo-panels, and weather-shock interactions to distinguish composition accounting, cohort structure, and a more direct vulnerability mechanism.

The main results are organized around the three portfolio margins. First, aging does not explain aggregate residential electrification. From 2000–2004 to 2020–2024, residential energy expenditure became more electricity-intensive in the CEX: the electricity share increased by about 5.1 percentage points, and the fossil home-energy share declined by the same amount. A DFL reweighting exercise (DiNardo et al., 1996) shows that age composition explains a -0.3 percentage point contribution to the electricity-share change, or roughly -6 percent of the observed increase, in the age-only specification. This residential fuel-mix result is not robust after incorporating geographic and housing-related composition.

Second, aging has a clearer relationship with transport fuel reliance. The gasoline share of residential plus transport energy expenditure rose by 2.09 percentage points over the sample comparison period, but the age-composition effect is -0.88 percentage points. This effect is economically meaningful relative to the observed change: an older age structure pushes the household energy portfolio away from gasoline even while other forces increased gasoline's expenditure share.

Third, aging is associated with greater vulnerability to heating shocks. RECS cross-sections show that elderly households have higher heating-cost shares and higher fossil home-energy shares. Weather-shock regressions provide the most direct mechanism evidence: a one-standard-deviation increase in state-year heating-degree-day shocks raises elderly households' log home-energy expenditure by an additional 0.95 percent and raises high-energy-burden risk by 0.87 percentage points. Income heterogeneity is consistent with affordability constraints amplifying the vulnerability margin, although burden outcomes mechanically depend on income. In contrast, fuel-mix responses to weather shocks are small and statistically insignificant, suggesting that the short-run mechanism is exposure to heating costs rather than immediate fuel switching. A cold/warm-state heterogeneity exercise further indicates that the burden response is concentrated in warmer states, consistent with weaker adaptation to unusually cold years.

Finally, a supplementary survival-adjusted shift-share IV check is reported in Appendix A. Because initial state age structure may be correlated with long-run housing, infrastructure, migration, and climate-adaptation patterns, the IV is not treated as the main identifying design.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes the conceptual framework. Section 4 introduces the data and variable construction. Section 5 presents the empirical methods. Section 6 reports the main results. Section 7 reports robustness checks and mechanism evidence. Section 8 discusses implications and limitations. Section 9 concludes.

## 2 Related Literature

This paper connects five strands of research. The first studies household energy demand and its demographic determinants. Residential energy use is shaped by income, prices, climate, building characteristics, appliance ownership, tenure, and household composition. Empirical work on residential energy demand shows that household demographics and housing characteristics are central to explaining heterogeneity in consumption and conservation behavior (Brounen et al., 2012; Rehdanz, 2007). The closest aging-specific contribution is Liao and Chang (2002), which studies space-heating and water-heating energy demand among older households in the United States. That paper highlights a key mechanism for this study: aging can matter not only through income but also through thermal comfort and time spent at home. The present analysis extends this line of work by examining a broader household energy portfolio that includes electricity, fossil home-energy fuels, gasoline, and energy-burden exposure.

The second strand studies life-cycle and cohort differences in consumption. Life-cycle models imply that household consumption bundles change with age as income, household size, employment, health, and mobility evolve (Browning et al., 1985). Energy consumption is especially likely to contain durable and cohort-specific components because heating systems, appliances, and housing structures are long-lived. In repeated cross-sections, this creates a difficult age-period-cohort problem: older households may differ because they are older, because they belong to cohorts with different housing and technology histories, or because they are observed in different energy-price and policy environments. Deaton (1985) provides a pseudo-panel framework for following cohorts in repeated cross-sections, while Costa and Kahn (2011) emphasizes that durable housing and cohort effects can shape electricity consumption. This paper therefore does not treat raw elderly/non-elderly differences as causal age effects. Instead, it combines cohort pseudo-panels with same-cohort aging changes to examine which patterns survive when cohort structure is made explicit.

The third strand examines weather, climate, adaptation, and energy demand. Heating and cooling degree days are widely used to measure weather-driven energy demand, and the empirical literature increasingly exploits weather variation to estimate climate and energy-consumption responses (Auffhammer and Mansur, 2014). Weather variation is also informative for vulnerability because it creates short-run demand shocks that households may have limited ability to avoid. Bhattacharya et al. (2003) shows that cold-weather shocks can force difficult tradeoffs among low-income households, while Deschênes and Greenstone (2011) emphasizes the importance of adaptation to climate exposure. In the analysis below, heating-degree-day shocks are used as a mechanism test: if aging increases heating-related exposure, elderly households are expected to experience larger home-energy and energy-burden responses when heating demand unexpectedly rises.

The fourth strand concerns energy burden and energy insecurity. Energy insecurity links energy costs, housing quality, health, and vulnerability to climate stress (Hernández, 2013, 2016; Jessel et al., 2019). Older households are particularly relevant in this framework because they may face fixed incomes, higher thermal sensitivity, reduced mobility, and greater health risks from under-heating or under-cooling. The energy-insecurity literature motivates looking beyond average energy expenditure to the risk of high burden and exposure to weather shocks.

The fifth strand concerns transport energy demand and household vehicle use. Household gasoline demand varies systematically with income, location, vehicle holdings, and travel needs (Schmalensee and Stoker, 1999; Small and Van Dender, 2007). Travel-survey evidence also shows that travel behavior differs strongly by age, household structure, and urban form (Pucher and Renne, 2003). Policies and prices that affect gasoline use have distributional consequences across households with different travel requirements (Bento et al., 2009). This literature motivates treating gasoline as part of the household energy portfolio rather than as a separate transportation-only outcome. Retirement, health, household composition, and reduced commuting can lower gasoline reliance even if residential energy needs rise. A household-level energy measure that includes only residential fuels can therefore miss a central channel through which aging changes energy demand.

The contribution of this paper is to bring these strands together in a single empirical setting. Rather than asking only whether older households consume more or less energy, the analysis distinguishes residential fuel mix, transport fuel reliance, and vulnerability to heating shocks. It also distinguishes expenditure shares from price-adjusted expenditure indexes. This distinction is important because changes in natural gas or fuel-oil prices can alter expenditure shares without necessarily reflecting physical electrification. The resulting evidence shows that aging is not the main driver of aggregate residential electrification, but it is relevant for gasoline demand and heating-related vulnerability.

# 3 Conceptual Framework

Population aging can affect the household energy portfolio through three channels. The framework is organized around residential energy demand, transport fuel demand, and energy burden.

First, residential energy demand can be written conceptually as:

$$\begin{aligned} ResidentialEnergy_{it} = {} & TimeAtHome_{it} \times ThermalComfort_{it} \\ & \times HousingEfficiency_{it}^{-1} \times FuelTechnology_{it}. \end{aligned}$$

Older individuals may spend more time at home and may have greater sensitivity to indoor temperature. If older households occupy larger or less efficient homes, the same comfort demand can translate into higher energy expenditure. Fuel technology and housing efficiency change slowly, so residential fuel-mix differences may reflect legacy heating systems and housing stock as much as contemporaneous age.

Second, transport fuel demand can be summarized as:

$$GasolineDemand_{it} = TripFrequency_{it} \times TripDistance_{it} \times VehicleEfficiency_{it}^{-1}.$$

Retirement and lower commuting needs can reduce trip frequency and distance. This mechanism works in the opposite direction from the residential heating channel: aging may raise exposure to home-energy costs while lowering gasoline reliance.

Third, energy burden depends jointly on prices, demand, and income:

$$EnergyBurden_{it} = \frac{EnergyPrice_{st} \times EnergyDemand_{it}}{Income_{it}}.$$

Older households may be more exposed to this margin if thermal comfort needs are less flexible, if incomes are fixed, or if housing systems make short-run adjustment difficult. Heating-degree-day shocks therefore provide a mechanism test for whether elderly households are more vulnerable when heating demand unexpectedly rises.

These channels motivate three empirical predictions. Elderly households are expected to have more heating-intensive home-energy bundles, but residential fuel-mix effects may be muted once cohort and housing-stock differences are taken into account. Elderly households are expected to rely less on gasoline because of life-cycle changes in mobility and commuting. Finally, elderly households are expected to experience larger increases in home-energy expenditure and burden during heating-demand shocks.

The data do not observe every component in the conceptual expressions. CEX contains expenditures, demographics, tenure, region, and state, but not time at home, physical energy quantities, detailed vehicle holdings, or housing efficiency. RECS provides richer residential-energy and housing information, including heating cost shares, home size, tenure, housing type, and fuel shares, but only for selected cross-sections. The empirical design therefore maps the framework to observable margins: residential fuel shares proxy for fuel-technology and housing-stock constraints; gasoline expenditure and gasoline participation proxy for transport fuel reliance; and weather-shock interactions test whether a plausibly exogenous increase in heating demand translates into larger burden responses for elderly households. Unobserved housing efficiency, time use, and vehicle efficiency remain mechanisms that are interpreted indirectly rather than separately identified.

# 4 Data and Measurement

## 4.1 Consumer Expenditure Survey

The main long-run analysis uses the CEX Interview public-use microdata from 2000 to 2024 (U.S. Bureau of Labor Statistics, 2026). The CEX provides repeated cross-sections of household expenditures, demographic characteristics, tenure, income, region, and state. Annual household measures of residential energy expenditure are constructed for electricity, natural gas, and fuel oil. Gasoline expenditure is also measured to capture transport fuel demand.

The main residential fuel-mix outcomes are:

$$\text{Electricity Share}_{it} = \frac{\text{Electricity Expenditure}_{it}}{\text{Electricity}_{it} + \text{Natural Gas}_{it} + \text{Fuel Oil}_{it}},$$

and

$$\text{Fossil Home Share}_{it} = \frac{\text{Natural Gas}_{it} + \text{Fuel Oil}_{it}}{\text{Electricity}_{it} + \text{Natural Gas}_{it} + \text{Fuel Oil}_{it}}.$$

A household is defined as elderly when the reference person is age 65 or older. Energy burden is measured as home-energy expenditure divided by income, winsorized at the 99th percentile in robustness checks. The main high-energy-burden indicator equals one when home-energy expenditure is at least 6 percent of household income. Alternative thresholds of 4, 8, and 10 percent are used in robustness checks.

## 4.2 Residential Energy Consumption Survey

The RECS microdata provide detailed information on residential energy use, heating costs, fuel mix, and energy insecurity (U.S. Energy Information Administration, 2026a). The 2015 and 2020 RECS waves are used to document cross-sectional differences between elderly and non-elderly households. RECS is especially useful for measuring heating expenditure and heating shares, which are central to the mechanism proposed in this paper.

## 4.3 Weather, Prices, and Population Data

State-year heating-degree-day and cooling-degree-day indicators are drawn from EIA state energy indicators (U.S. Energy Information Administration, 2026b). Deviations from state-specific weather norms are used to construct heating-degree-day shocks. EIA SEDS state-year fuel prices are used to construct price-adjusted expenditure indexes. The matched prices cover residential electricity, natural gas, distillate fuel oil, and motor gasoline, all measured in dollars per million Btu. Census state age distributions from 2000 to 2024 are used to measure actual state-year elderly shares and to construct a predetermined age-structure instrument (U.S. Census Bureau, 2026).

## 4.4 Measurement Distinctions

The analysis uses several related but distinct measures of aging. At the household level, the main CEX measure is an elderly-reference household, defined by whether the household reference person is age 65 or older. This variable is best interpreted as a household life-cycle measure. RECS elderly-household status is used for cross-sectional mechanism evidence and follows the classification available in that survey. At the state-year level, Census elderly shares measure population age structure rather than household life-cycle status. The survival-adjusted IV focuses on the Census 65–74 population share and is therefore narrower than the household-level elderly indicator. These distinctions matter for interpretation: household regressions speak to elderly-household differences, while state-level exercises speak to population-composition changes.

## 4.5 Variable Definitions

Table 1 summarizes the key variables. The main outcomes are based on expenditures because the long CEX public-use files do not report physical energy quantities. Accordingly, fuel-mix outcomes are interpreted as expenditure shares, not physical energy shares. Price-adjusted variables are described as expenditure indexes rather than metered quantities because they divide household expenditures by state-year average fuel prices. They reduce the role of relative price movements but do not recover household-level physical consumption.

Table 1: Variable Definitions

| Variable | Definition | Source |
|---|---|---|
| Elderly household | Indicator equal to one if the household reference person is age 65 or older | CEX |
| Any elderly household | Indicator equal to one if any household member is age 65 or older | CEX |
| Home energy expenditure | Annual spending on electricity, natural gas, and fuel oil | CEX |
| Electricity share | Electricity expenditure divided by electricity, natural gas, and fuel-oil expenditure | CEX |
| Fossil home-energy share | Natural-gas plus fuel-oil expenditure divided by home-energy expenditure | CEX |
| Gasoline share | Gasoline expenditure divided by residential energy plus gasoline expenditure | CEX |
| Price-adjusted electricity index | Electricity expenditure divided by state-year electricity price, as a share of home-energy expenditure index | CEX, EIA |
| Price-adjusted fossil index | Natural-gas and fuel-oil expenditure divided by matched prices, as a share of home-energy expenditure index | CEX, EIA |
| Energy burden | Home-energy expenditure divided by household income | CEX |
| High energy burden | Indicator for home-energy expenditure at least 6 percent of income | CEX |
| Low-income household | Indicator for the bottom tercile of year-specific equivalized household income | CEX |
| Heating share | Space-heating cost divided by total home-energy cost | RECS |
| HDD shock | Within-state standardized deviation of annual heating degree days from state norms | EIA |
| Elderly share | Census 65+ population share in a state-year | Census |
| Predicted 65–74 share | Survival-adjusted predicted 65–74 share based on year-2000 state age structure | Census |

Notes: CEX outcomes are expenditure-based. RECS variables are used for cross-sectional mechanism evidence on heating intensity and energy insecurity. Price-adjusted indexes divide expenditures by state-year average prices and should not be interpreted as household metered consumption. The predicted elderly share is predetermined by year-2000 state age structure and calendar-time cohort aging.

# 5 Empirical Strategy

The empirical strategy is not designed around a single sufficient causal estimator. Instead, it uses a layered design in which each component addresses a different part of the research question. DFL reweighting provides composition accounting for long-run portfolio changes. Cohort pseudo-panels examine whether elderly-household differences are absorbed by cohort structure. Weather-shock interactions provide the most direct mechanism test for heating-related vulnerability because annual weather deviations are plausibly exogenous conditional on state and year fixed effects. The shift-share IV is reported in the appendix as a supplementary state-level check and is interpreted cautiously because its exclusion restriction is stronger than that required for the weather-shock design.

## 5.1 DFL Reweighting

The first empirical exercise asks how much of the long-run change in the household energy portfolio can be attributed to changes in household age composition. The analysis compares an early period, 2000–2004, with a late period, 2020–2024. Following DiNardo et al. (1996), late-period households are reweighted so that their observable characteristics match the early-period distribution. The counterfactual mean for outcome $Y$ is:

$$\bar{Y}_{late}^{CF} = \frac{\sum_{i \in late} w_i \psi(X_i) Y_i}{\sum_{i \in late} w_i \psi(X_i)},$$

where $w_i$ is the survey weight and $\psi(X_i)$ is the reweighting factor based on age or an expanded set of household characteristics. In practice, $\psi(X_i)$ is estimated from a weighted logit model for belonging to the late period. The age-only specification uses CEX age groups. Expanded specifications add household size, region, tenure, urban status, and state. The composition effect is:

$$\Delta^{comp} = \bar{Y}_{late} - \bar{Y}_{late}^{CF}.$$

This measures how much of the observed change is associated with the change in the distribution of aging-related characteristics. It is a decomposition exercise, not a causal estimate of biological aging.

## 5.2 Cohort Pseudo-Panels

Repeated cross-sections make it difficult to distinguish aging from cohort habits. To address this concern, birth-cohort pseudo-panels are constructed following Deaton (1985). Households are grouped into cohort-year cells, and outcomes are related to age profiles while absorbing period and cohort variation. This design does not fully resolve the age-period-cohort identification problem, but it checks whether the raw elderly differences are driven entirely by fixed cohort differences. The pseudo-panel evidence is therefore interpreted as a cohort-adjustment exercise rather than as a stand-alone causal design.

## 5.3 Weather-Shock Mechanism Test

The weather-shock specification tests whether elderly households are differentially exposed to heating-demand shocks:

$$Y_{ist} = \beta \left(HDDShock_{st} \times Elderly_i\right) + \theta HDDShock_{st} + \gamma Elderly_i + X_{ist}'\delta + \alpha_s + \lambda_t + \varepsilon_{ist}.$$

Here $Y_{ist}$ is an energy outcome for household $i$ in state $s$ and year $t$, $HDDShock_{st}$ is the within-state standardized deviation of annual heating degree days from the state mean, $\alpha_s$ are state fixed effects, and $\lambda_t$ are year fixed effects. A one-unit increase in $HDDShock_{st}$ corresponds to a one-standard-deviation colder-than-usual year within a state. A positive $\beta$ for home-energy expenditure or energy burden indicates that elderly households are more vulnerable when heating demand unexpectedly rises.

To examine whether this vulnerability is concentrated among economically constrained older households, the analysis also estimates a compact income-heterogeneity specification:

$$
\begin{aligned}
Y_{ist} = {} & \beta_1 HDDShock_{st} + \beta_2 \left(HDDShock_{st} \times Elderly_i\right) \\
& + \beta_3 \left(HDDShock_{st} \times LowIncome_i\right) + \beta_4 \left(HDDShock_{st} \times Elderly_i \times LowIncome_i\right) \\
& + X'_{ist}\delta + \alpha_s + \lambda_t + \varepsilon_{ist}.
\end{aligned}
$$

The main low-income indicator equals one for households in the bottom tercile of year-specific equivalized household income, where income is divided by the square root of household size. The coefficient $\beta_4$ measures whether low-income elderly households have a larger heating-shock response than other elderly households. For expenditure outcomes, this is interpreted as differential spending response; for energy-burden outcomes, it captures affordability consequences as well as spending responses because income enters the denominator.

## 5.4 Shift-Share IV

As a supplementary state-level check, the analysis constructs a predetermined state age-structure instrument. For each state, the year-2000 age distribution is mechanically aged forward. The preferred survival-adjusted version uses national cohort progression to predict the 65–74 population share in each state-year. The second-stage state-year specification is:

$$
Y_{st} = \rho AgingShare_{st} + Z'_{st}\eta + \alpha_s + \lambda_t + u_{st}.
$$

The IV is reported cautiously for two reasons. First, the preferred version identifies predicted entry into ages 65–74 rather than the full elderly population stock. Second, initial state age structure may be correlated with long-run housing stock, urban form, regional energy infrastructure, retirement migration, and climate adaptation. State and year fixed effects absorb time-invariant state differences and common national shocks, but they do not by themselves rule out differential state trends correlated with baseline age structure. For this reason, the IV evidence is used to assess consistency with the household-level findings, not to replace the weather-shock mechanism as the main source of identification.

# 6 Results

## 6.1 Aggregate Trends

Figure 1 shows the long-run CEX trends. Residential energy expenditure became more electricity-intensive between 2000 and 2024. At the same time, the share of elderly-reference households rose. These aggregate trends motivate the central decomposition exercise: how much of the observed household energy-portfolio change is associated with age composition, and how much reflects other period-specific forces?

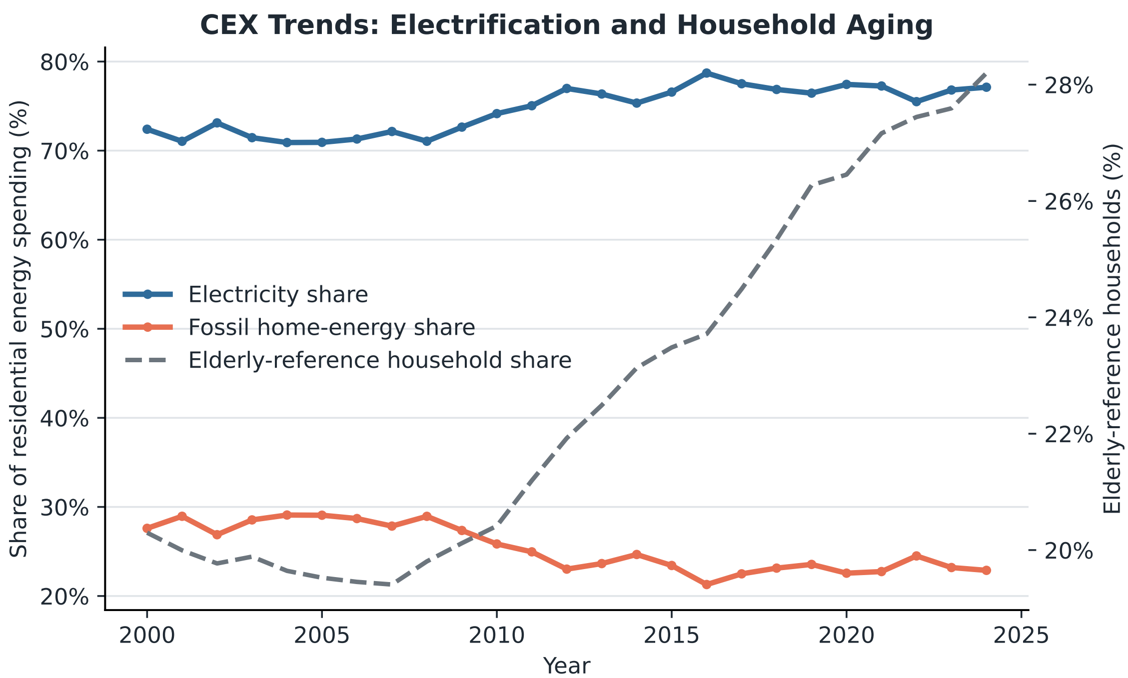


Figure 1: CEX Trends in the Household Energy Portfolio and Aging

## 6.2 DFL Decomposition

Table 2 summarizes the main DFL decomposition. The electricity share rose by 5.06 percentage points from 2000–2004 to 2020–2024. In the age-only reweighting specification, aging explains a -0.30 percentage point contribution to the electricity-share change. This is about -6 percent of the observed increase. Equivalently, aging raises the fossil home-energy share by 0.30 percentage points. This age-only result suggests a small offset to electrification, but Table 3 shows that the residential fuel-mix effect is not robust once geographic and housing-related composition is incorporated.

The largest age-related composition effect appears in gasoline reliance. The gasoline share of residential plus transport energy expenditure rose by 2.09 percentage points overall, but the age-only composition effect is -0.88 percentage points. In magnitude, this offset is about 42 percent of the observed gasoline-share increase. Thus, the changing age structure reduces transport fuel reliance relative to what would have occurred under a younger demographic structure. The DFL results therefore support the portfolio interpretation: age composition has little explanatory power for residential electrification, but it matters more for the allocation between residential and transport energy.

Table 2: DFL Decomposition of CEX Energy-Portfolio Outcomes, 2000–2004 vs. 2020–2024

| Outcome | Early Mean | Late Mean | Observed Change | Age Composition |
|---|---|---|---|---|
| Home energy expenditure ($) | 813.12 | 1242.51 | 429.39 | 3.88 |
| Electricity share (p.p.) | 71.77 | 76.84 | 5.06 | -0.30 |
| Fossil home-energy share (p.p.) | 28.23 | 23.16 | -5.06 | 0.30 |
| Gasoline share (p.p.) | 42.34 | 44.44 | 2.09 | -0.88 |
| High energy burden (p.p.) | 15.25 | 13.22 | -2.02 | 0.87 |

Notes: Shares are reported in percentage points. The age-composition effect is the difference between the late observed mean and the late counterfactual mean reweighted to the early-period age distribution. Positive values indicate that aging raises the outcome relative to the counterfactual.

Table 3 reports the age-composition effect across alternative reweighting specifications. The residential fuel-mix effect is sensitive to the added controls, which is consistent with the interpretation that housing, tenure, region, and state composition are important for residential fuel shares. The gasoline result is more stable: the age-composition effect remains negative across all specifications, ranging from -0.47 to -0.88 percentage points. High-burden effects are also positive across specifications, although their magnitude varies with the control set.

Table 3: DFL Age-Composition Effects Across Reweighting Specifications

| Outcome | Age Only | Age + HH Size | Age + Region/Tenure/Urban | Age + State/Tenure/Urban |
|---|---|---|---|---|
| Electricity share | -0.30 | -0.37 | 0.44 | 0.73 |
| Fossil home-energy share | 0.30 | 0.37 | -0.44 | -0.73 |
| Gasoline share | -0.88 | -0.79 | -0.58 | -0.47 |
| High energy burden | 0.87 | 0.84 | 0.46 | 1.51 |

Notes: Entries are DFL composition effects in percentage points. The age-only specification uses CEX age groups. The expanded specifications add household size and then region, tenure, urban status, and state indicators. Positive values indicate that late-period age-related composition raises the outcome relative to the counterfactual late-period distribution reweighted to early-period covariates.

Because the gasoline-share result could mechanically reflect changes in the denominator, Table 4 reports gasoline components directly. Elderly households spend substantially less on gasoline in both periods, have lower log gasoline expenditure conditional on positive spending, and are less likely to report positive gasoline spending. Home-energy expenditure is similar across elderly and non-elderly households within each period. The lower gasoline share is therefore not only a denominator effect from higher residential energy expenditure; it mainly reflects lower transport fuel spending and lower gasoline participation among elderly households.

Table 4: Gasoline and Home-Energy Components by Elderly Status

| Period | Group | Gasoline Expenditure | Log Gasoline > 0 | Any Gasoline | Home Energy | Gasoline Share |
|---|---|---|---|---|---|---|
| 2000–2004 | Non-elderly | 762.37 | 6.45 | 92.83 | 818.81 | 45.16 |
| 2000–2004 | Elderly | 419.15 | 5.87 | 84.69 | 789.50 | 31.02 |
| 2020–2024 | Non-elderly | 1426.82 | 7.01 | 92.70 | 1251.00 | 47.58 |
| 2020–2024 | Elderly | 847.62 | 6.48 | 89.22 | 1220.01 | 36.10 |

Notes: CEX weighted means. Gasoline and home-energy expenditure are in nominal dollars. "Log Gasoline $>$ 0" is the weighted mean of log gasoline expenditure among households with positive gasoline spending. Any gasoline and gasoline share are reported in percentage points.

Figure 2 visualizes the age-only decomposition. Aging is not large enough to explain the aggregate shift toward electricity, but it pushes the household portfolio toward higher fossil residential share and lower gasoline share.

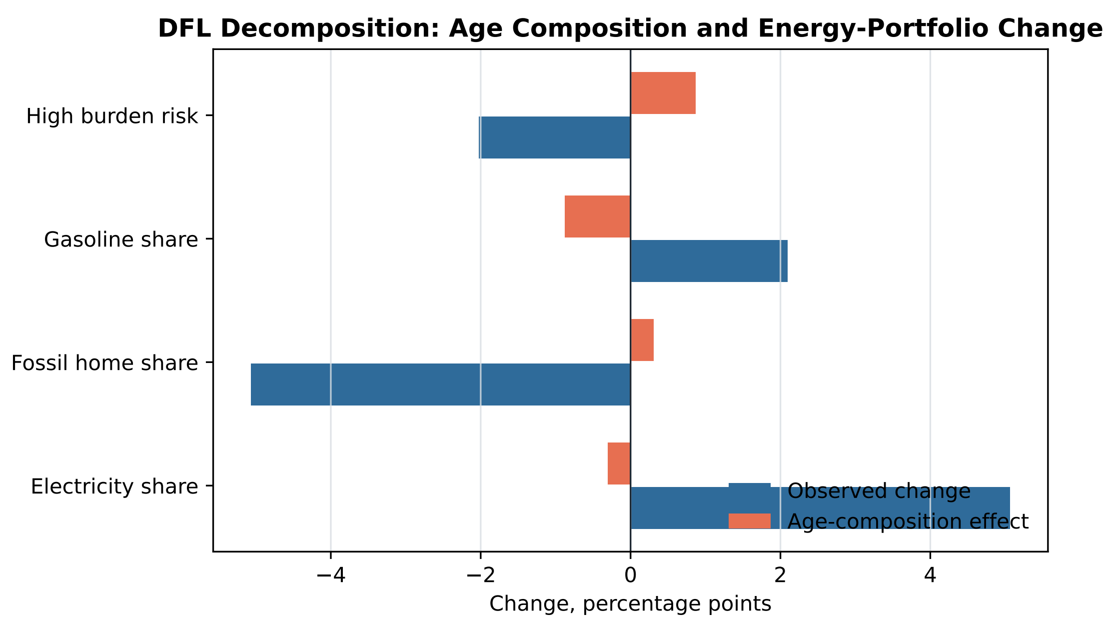


Figure 2: DFL Decomposition of Energy-Portfolio Changes

### 6.3 RECS Cross-Sectional Evidence

RECS provides a more detailed view of heating-related residential energy use. Table 5 shows that elderly households have higher heating shares and higher fossil home-energy shares in both 2015 and 2020. In 2020, for example, elderly households have a fossil home-energy share of 28.3 percent, compared with 24.1 percent for non-elderly households. Their heating share is also about 5.4 percentage points higher.

Table 5: RECS Elderly and Non-Elderly Household Energy Profiles

| Year | Group | Home Energy | Heating Share | Electricity Share | Fossil Share |
|---|---|---|---|---|---|
| 2015 | Non-elderly | 1860.18 | 24.49 | 75.67 | 24.33 |
| 2015 | Elderly | 1857.06 | 29.38 | 73.65 | 26.35 |
| 2020 | Non-elderly | 1897.40 | 23.24 | 75.91 | 24.09 |
| 2020 | Elderly | 1856.75 | 28.60 | 71.70 | 28.30 |

Notes: RECS household-weighted means. Heating, electricity, and fossil shares are reported in percentage points. Elderly households are defined by elderly household status in the RECS classification used in the analysis.

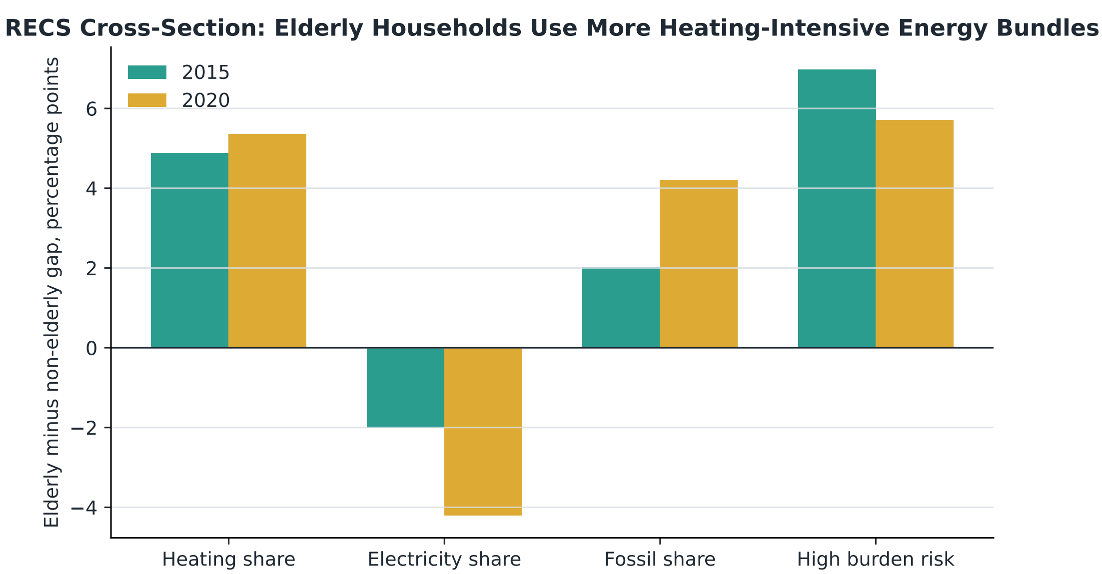


Figure 3: Elderly–Non-Elderly Gaps in RECS Energy Outcomes

# 7 Robustness and Mechanisms

## 7.1 Price-Adjusted Expenditure Indexes

The main CEX outcomes are expenditure shares, so they can change mechanically when relative fuel prices change. This is especially relevant for natural gas, whose price movements can lower the fossil home-energy expenditure share even if physical gas use does not decline. To address this concern, EIA SEDS state-year prices are merged to the CEX, and fuel-specific price-adjusted expenditure indexes are constructed by dividing CEX expenditures by matched fuel prices in dollars per million Btu. The resulting variables are not interpreted as metered household consumption, but they provide a direct robustness check for whether the expenditure-share patterns are driven only by relative price movements.

Table 6 compares the expenditure-share results with price-adjusted expenditure-index shares. The expenditure-based electricity share rises by 6.3 percentage points between 2000–2004 and 2019–2023. The price-adjusted electricity index also rises, by 5.0 percentage points. The age-composition effect remains small and negative for electricity share and positive for fossil home-energy share. Thus, the residential fuel-mix conclusion is not fully explained by natural gas or fuel-oil price movements. At the same time, this exercise should be read as a price-adjustment check, not as a substitute for household-level physical energy quantities.

Table 6: Price-Adjusted Expenditure-Index Robustness

| Outcome | Early Mean | Late Mean | Observed Change | Age Composition |
|---|---|---|---|---|
| Expenditure electricity share (p.p.) | 70.28 | 76.55 | 6.27 | -0.28 |
| Price-adjusted electricity index (p.p.) | 57.64 | 62.67 | 5.02 | -0.27 |
| Expenditure fossil home share (p.p.) | 29.72 | 23.45 | -6.27 | 0.28 |
| Price-adjusted fossil home index (p.p.) | 42.36 | 37.33 | -5.02 | 0.27 |
| Expenditure gasoline share (p.p.) | 41.98 | 44.19 | 2.21 | -0.89 |
| Price-adjusted gasoline index (p.p.) | 49.48 | 47.57 | -1.91 | -0.88 |

Notes: Price-adjusted outcomes divide CEX fuel-specific expenditures by EIA SEDS state-year prices in dollars per million Btu. The late period is 2019–2023 because EIA SEDS prices are currently available through 2023. These variables are price-adjusted expenditure indexes, not metered household quantities. The age-composition effect is from the age-only DFL reweighting.

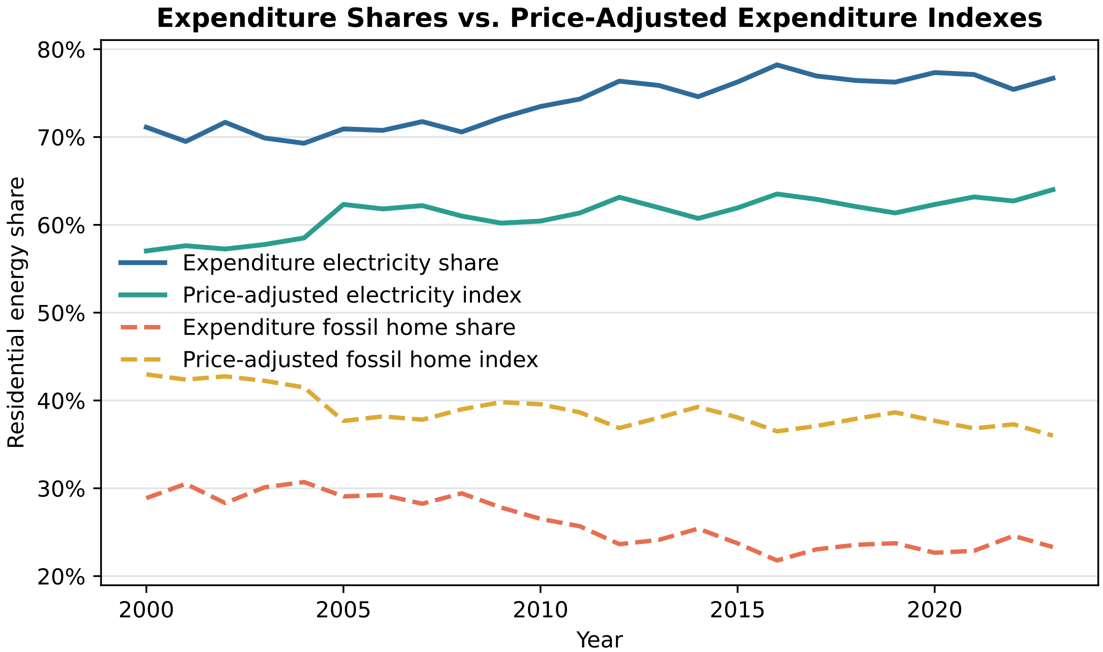


Figure 4: Expenditure Shares and Price-Adjusted Expenditure Indexes

## 7.2 Cohort and Pseudo-Panel Checks

A central concern is that elderly households may differ from younger households because they belong to different birth cohorts, not because of aging itself. The cohort pseudo-panel results indicate that cohort differences matter, especially for residential fuel mix. After controlling for cohort patterns, the elderly fuel-mix gap becomes weaker than in raw comparisons. However, the gasoline result remains stable: older households consistently rely less on gasoline. This supports the interpretation that the transport-fuel effect is a life-cycle pattern, while residential fuel mix is partly shaped by cohort and housing-stock inertia.

Table 7 reports selected pseudo-panel estimates from cohort-year cells. The preferred specification includes cohort and year fixed effects and household controls. The age profile for electricity and fossil home-energy shares becomes statistically imprecise after these controls are added, while the gasoline share declines more steeply with age at older ages. This pattern is consistent with the interpretation that residential fuel mix partly reflects cohort and housing-stock differences, whereas lower gasoline reliance is more consistently associated with aging.

Table 7: Cohort Pseudo-Panel Age-Profile Estimates

| Outcome | Age | Age$^2$ | Slope at 65 | Slope at 75 |
|---|---|---|---|---|
| Electricity share | -0.0022 | 0.000004 | -0.0020 | -0.0020 |
| | (0.0015) | (0.000006) | | |
| Fossil home-energy share | 0.0022 | -0.000004 | 0.0020 | 0.0020 |
| | (0.0015) | (0.000006) | | |
| Home energy expenditure | 28.0358*** | -0.1073*** | 24.8178 | 22.6725 |
| | (4.7958) | (0.0178) | | |
| High energy burden | 0.0049** | -0.000058*** | 0.0032 | 0.0020 |
| | (0.0020) | (0.000007) | | |
| Gasoline share | -0.0017 | -0.000104*** | -0.0048 | -0.0069 |
| | (0.0013) | (0.000005) | | |

Notes: Standard errors are in parentheses. Estimates are based on CEX birth-cohort-year pseudo-panel cells. The displayed specification includes cohort fixed effects, year fixed effects, region fixed effects, average household size, average log income, owner share, and urban share. Age is centered in the underlying regression. Slopes at ages 65 and 75 combine the linear and quadratic age terms. Share outcomes are in unit shares, not percentage points. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As a more transparent complement to the APC-style pseudo-panel, the analysis also follows the same five-year birth-cohort-by-region cells as they move across age stages. Before computing these within-cohort changes, common year shocks are removed by subtracting weighted year means. Figure 5 reports the year-residualized changes. The gasoline share declines as the same cohorts move from ages 55–64 to 65–74 and again from 65–74 to 75–84. High energy burden rises as cohorts enter older ages. Residential fuel-mix changes are smaller, with fossil home-energy shares rising modestly. These patterns do not eliminate the APC identification problem, but they support the interpretation that lower gasoline reliance and higher energy vulnerability are not only fixed cohort habits.

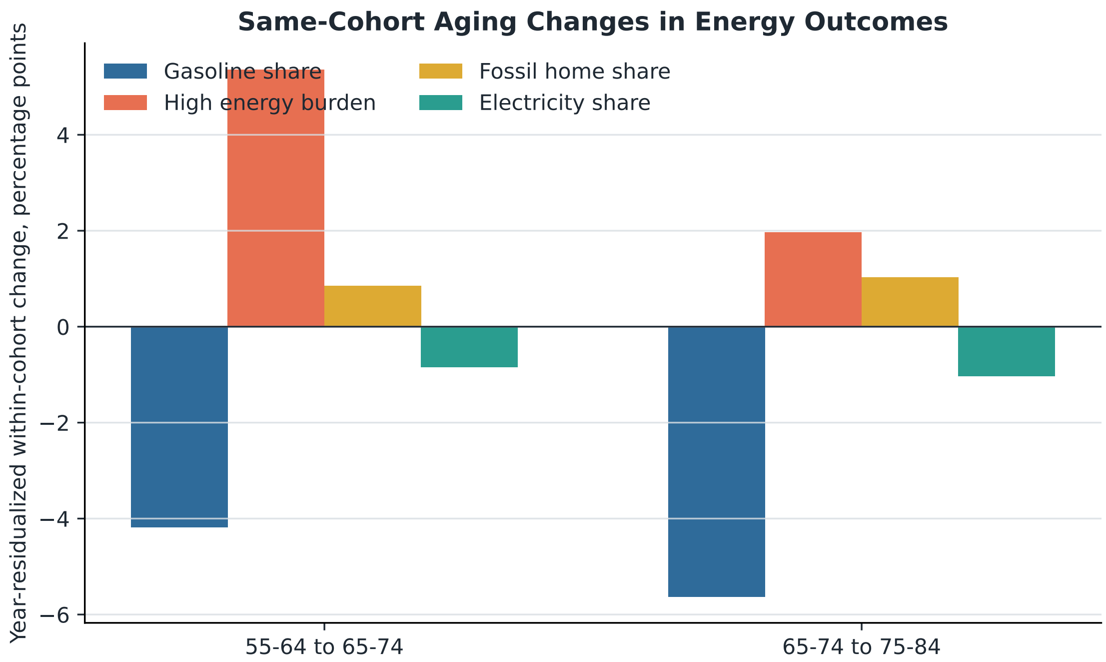


Figure 5: Same-Cohort Aging Changes in Energy Outcomes

## 7.3 Weather-Shock Evidence

Table 8 reports the household-level weather-shock interaction results. The HDD shock is standardized within state, so coefficients measure responses to a one-standard-deviation colder-than-usual year. Non-elderly households increase log home-energy expenditure by about 1.99 percent in such years; elderly households have an additional 0.95 percent response, implying a total response of about 2.94 percent. For high energy burden, the non-elderly main effect is 0.33 percentage points, while the elderly interaction is 0.87 percentage points, implying a total elderly response of 1.19 percentage points. Coefficients for electricity and fossil residential shares are small and statistically insignificant.

Table 8: Weather-Shock Mechanism: HDD Main Effects and Elderly Interactions

| Outcome | HDD Main Effect | HDD × Elderly | Elderly Total Response |
|---|---|---|---|
| Log home energy | 0.0199*** | 0.0095*** | 0.0294 |
| | (0.0051) | (0.0036) | |
| Home energy expenditure | 19.1228*** | 9.2550** | 28.3778 |
| | (6.8381) | (4.1932) | |
| High energy burden | 0.0033** | 0.0087*** | 0.0119 |
| | (0.0014) | (0.0024) | |
| Energy burden, winsorized | 0.0009 | 0.0030** | 0.0038 |
| | (0.0007) | (0.0013) | |
| Electricity share | -0.0007 | -0.0008 | -0.0015 |
| | (0.0014) | (0.0010) | |
| Fossil home-energy share | 0.0007 | 0.0008 | 0.0015 |
| | (0.0014) | (0.0010) | |
| Gasoline share | -0.0027** | 0.0003 | -0.0023 |
| | (0.0012) | (0.0014) | |

Notes: Standard errors are in parentheses. Coefficients are from household-level CEX regressions with HDD shock interacted with elderly-reference household status. HDD shock is standardized within state; a one-unit increase corresponds to a one-standard-deviation colder-than-usual year. The main effect is the response for non-elderly-reference households. The elderly total response is the sum of the main effect and the interaction. Specifications include household controls, state fixed effects, year fixed effects, CDD-shock interactions, survey weights, and state-clustered standard errors. $N = 495{,}373$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The high-burden result is not specific to the 6 percent threshold. Table 9 shows that the elderly interaction remains positive and statistically significant when high burden is defined using 4, 8, or 10 percent of income.

Table 9: Weather-Shock Robustness Across Energy-Burden Thresholds

| High-Burden Threshold | HDD × Elderly | p-value |
|---|---|---|
| 4 percent | 0.0076*** | 0.000 |
| 6 percent | 0.0087*** | 0.000 |
| 8 percent | 0.0072*** | 0.005 |
| 10 percent | 0.0056** | 0.012 |

Notes: Entries report the coefficient on HDD shock interacted with elderly-reference household status from the same household-level weather-shock specification as Table 8. Each row changes the high-energy-burden threshold. Standard errors are clustered at the state level.

Figure 6 shows the baseline weather-shock evidence graphically. The results are consistent with the heating-vulnerability mechanism. Elderly households are more exposed to heating shocks in expenditure and burden terms, while short-run fuel-mix responses remain limited. This finding is central to the paper's interpretation because it links aging to vulnerability under an externally driven demand shock rather than only to cross-sectional differences in spending.

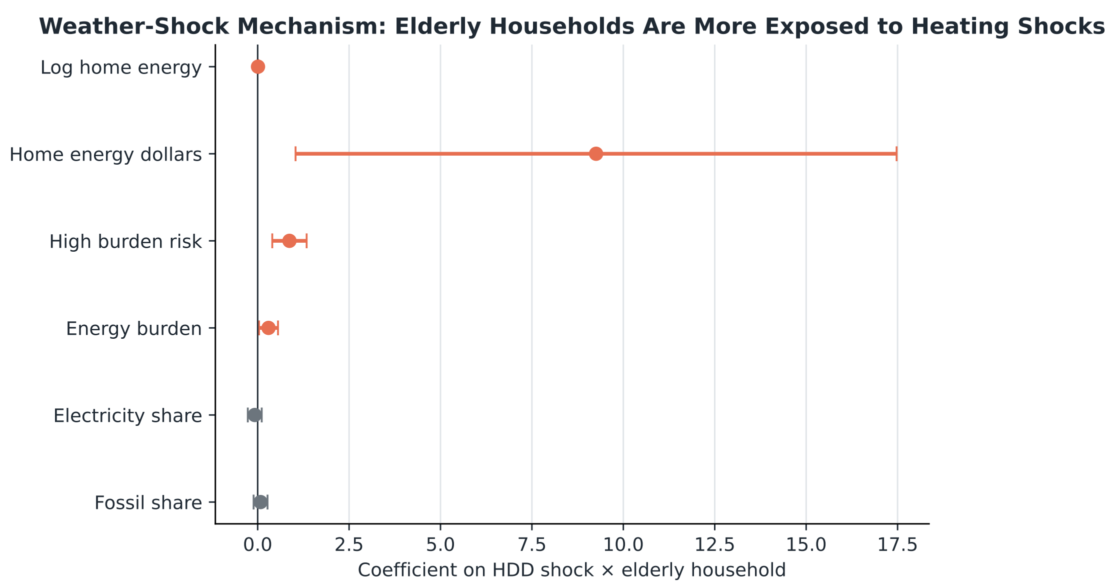


Figure 6: Weather-Shock Interaction Estimates

## 7.4 Income Heterogeneity in Heating-Shock Vulnerability

Table 10 asks whether the elderly heating-shock response is concentrated among low-income households. Low income is defined as the bottom tercile of year-specific equivalized household income. The results are consistent with affordability constraints amplifying the vulnerability margin. The triple interaction is positive for log home-energy expenditure, but only marginally significant. By contrast, the high-burden response is substantially larger for low-income elderly households: a one-standard-deviation colder year raises high-burden risk by about 2.13 percentage points for this group, compared with 0.39 percentage points for higher-income elderly households.

The negative coefficients on $HDDShock_{st} \times LowIncome_i$ for log and dollar home-energy expenditure suggest that low-income non-elderly households have weaker spending responses in colder-than-usual years, consistent with heating restraint or under-consumption. The positive elderly-low-income triple interaction partly offsets this pattern, suggesting that low-income elderly households may have less room to compress heating demand. For high-burden and winsorized-burden outcomes, larger responses also reflect the income denominator and should be interpreted as affordability consequences, not only as higher physical energy use. The triple-interaction coefficient for high burden remains positive under alternative low-income definitions, with estimates of 0.0070 for the bottom quartile and 0.0093 for below-median equivalized income.

Table 10: Income Heterogeneity in Heating-Shock Vulnerability

| Outcome | HDD Main | HDD × Elderly | HDD × Low Income | HDD × Elderly × Low Income | Higher-Income Elderly Total | Low-Income Elderly Total |
|---|---|---|---|---|---|---|
| Log home energy | 0.0237*** | 0.0065 | -0.0126*** | 0.0104* | 0.0302*** | 0.0280*** |
| | (0.0051) | (0.0042) | (0.0039) | (0.0059) | (0.0068) | (0.0075) |
| Home energy expenditure | 22.4882*** | 7.5225 | -10.8125*** | 7.2969 | 30.0106*** | 26.4951*** |
| | (6.9240) | (6.2393) | (4.1569) | (6.8959) | (8.6594) | (8.3626) |
| High energy burden | 0.0007 | 0.0032** | 0.0051 | 0.0123*** | 0.0039*** | 0.0213*** |
| | (0.0016) | (0.0013) | (0.0039) | (0.0048) | (0.0010) | (0.0050) |
| Energy burden, winsorized | -0.0004 | 0.0006*** | 0.0023 | 0.0060* | 0.0001 | 0.0084*** |
| | (0.0006) | (0.0002) | (0.0023) | (0.0031) | (0.0005) | (0.0030) |

Notes: Standard errors are in parentheses. Low income is defined as the bottom tercile of year-specific equivalized household income, where income is divided by the square root of household size. Total-response standard errors are computed by the delta method using the state-clustered covariance matrix. The higher-income elderly total is the sum of the HDD main effect and HDD × elderly. The low-income elderly total is the sum of the HDD main effect, HDD × elderly, HDD × low income, and the triple interaction. Specifications include household controls, state fixed effects, year fixed effects, CDD-shock interactions with the same elderly and low-income structure, survey weights, and state-clustered standard errors. $N = 495{,}373$. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

## 7.5 Climate-Zone Heterogeneity

The heating-vulnerability mechanism may differ across warmer and colder parts of the country. To examine this possibility, states are divided according to their sample-period average heating degree days. States with average HDD at or above the median are classified as cold states, and the remaining states are classified as warm states. This split uses the same EIA weather data as the baseline mechanism test and does not require additional external data.

Table 11 reports split-sample estimates of the coefficient on $HDDShock_{st} \times Elderly_i$. The results do not indicate that elderly-specific responses are larger in cold states. Instead, the high-burden response is larger in warm states: the coefficient is 0.0120 in warm states and 0.0034 in cold states, with a statistically significant cold-minus-warm difference. Log home-energy and dollar home-energy responses also appear larger in warm states, although the between-group differences are imprecisely estimated. Table 12 provides descriptive support for an adaptation interpretation: cold areas have higher routine heating shares, higher fossil home-energy shares, and greater fuel-oil and natural-gas reliance. These patterns are consistent with more extensive adaptation to regular heating demand in cold areas, whereas unusually cold years in warmer areas create more binding vulnerability shocks for older households.

Table 11: Climate-Zone Heterogeneity in Weather-Shock Responses

| Outcome | Warm States | Cold States | Cold Minus Warm |
|---|---|---|---|
| Log home energy | 0.0110** | 0.0062 | -0.0041 |
| | (0.0049) | (0.0047) | (0.0067) |
| Home energy expenditure | 9.6839** | 7.7781 | -0.8088 |
| | (4.5554) | (8.0854) | (9.1043) |
| High energy burden | 0.0120*** | 0.0034 | -0.0086** |
| | (0.0036) | (0.0022) | (0.0042) |
| Energy burden, winsorized | 0.0041* | 0.0013 | -0.0027 |
| | (0.0021) | (0.0011) | (0.0024) |

Notes: Standard errors are in parentheses. Entries report the coefficient on heating-degree-day shock interacted with elderly-reference household status. Warm and cold states are defined by whether a state's sample-period average HDD is below or above the median across states. Specifications include the same controls, state fixed effects, year fixed effects, CDD-shock interactions, survey weights, and state-clustered standard errors as the baseline weather-shock model. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: RECS Evidence on Cold- and Warm-Area Heating Mechanisms

| Climate Group | Group | Heating Share | Fossil Share | Fuel Oil Share | Natural Gas Share | Energy Insecurity | High Burden |
|---|---|---|---|---|---|---|---|
| Warm areas | Non-elderly | 17.71 | 17.35 | 0.33 | 15.53 | 36.50 | 23.02 |
| Warm areas | Elderly | 21.39 | 20.04 | 0.94 | 16.46 | 20.04 | 28.19 |
| Cold areas | Non-elderly | 31.46 | 32.66 | 3.81 | 25.66 | 30.63 | 21.53 |
| Cold areas | Elderly | 37.92 | 36.30 | 5.45 | 26.70 | 14.23 | 28.09 |

Notes: RECS weighted means from the 2015 and 2020 waves. Warm and cold areas are defined by whether household HDD65 is below or above the sample median. Entries are reported in percentage points. The table documents that cold areas have higher routine heating intensity and greater fossil-fuel heating reliance, consistent with more extensive adaptation to normal heating demand.

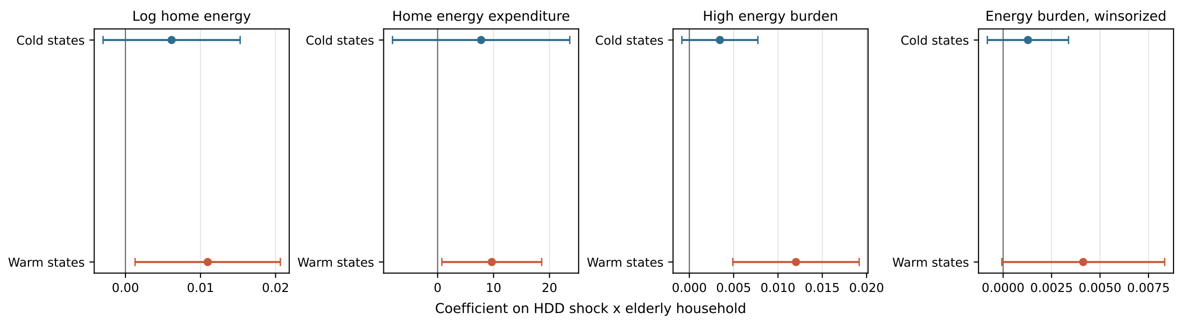


Figure 7: Cold- and Warm-State Heterogeneity in Weather-Shock Responses

Supplementary state-level IV checks are reported in Appendix A. They point in the same direction for gasoline reliance but are not used as the paper's central causal evidence because the exclusion restriction is demanding and the estimates for vulnerability outcomes are imprecise.

# 8 Discussion

The evidence suggests that aging affects the household energy portfolio in economically relevant ways, although it is not a dominant driver of aggregate residential electrification. This distinction sharpens the paper's main contribution. A narrow residential-electrification interpretation would emphasize a small and unstable fuel-mix effect. A broader portfolio interpretation shows a clearer pattern: aging shifts households away from gasoline reliance and increases exposure to heating-related energy burden, while residential fuel-mix differences are partly tied to slower-moving housing and cohort structure.

This distinction matters for policy. Policies aimed at electrification require attention to housing stock, regional fuel infrastructure, energy prices, and technology adoption rather than demographic aging alone. Policies aimed at energy affordability and climate adaptation should account for aging because elderly households appear more vulnerable to heating shocks, especially when old age is combined with low income. The cold/warm-state heterogeneity results further suggest that vulnerability is not simply a function of living in cold places. Warmer states may be less adapted to unusually cold years, making older households more exposed when such shocks occur.

Several limitations remain. First, CEX measures expenditures rather than physical energy quantities, so price variation can affect the measured energy structure. The price-adjusted expenditure index reduces this concern but does not replace metered household energy use. Second, repeated cross-sections cannot fully resolve the age-period-cohort identification problem. The cohort pseudo-panel and same-cohort aging checks reduce this concern but do not eliminate it. Third, the state-level IV strategy identifies predicted entry into ages 65–74, not the full elderly population stock, and its exclusion restriction is vulnerable to concerns about baseline age structure, housing stock, infrastructure, migration, and differential regional trends. Future work could strengthen identification using more granular geography, utility-service territories, housing-stock instruments, policy variation affecting heating technology adoption, or direct household energy-quantity data.

# 9 Conclusion

This paper studies how population aging is associated with the household energy portfolio in the United States. Using RECS, CEX, EIA weather indicators, EIA fuel prices, and Census age distributions, the analysis finds that aging has no robust explanatory role for residential electrification, reduces gasoline reliance, and increases elderly households' vulnerability to heating-related energy shocks. The most direct mechanism evidence comes from weather-shock interactions: older households experience larger increases in home-energy expenditure and energy-burden risk when heating demand unexpectedly rises, and the burden response is amplified among low-income elderly households.

The broader conclusion is that population aging belongs in energy-demand and energy-burden analysis, but the relevant margin of adjustment matters. Aging is not the main engine of aggregate residential electrification trends. Rather, it reallocates the household energy portfolio away from gasoline and toward greater home-energy vulnerability. Residential-energy-only measures can therefore misstate the energy implications of demographic change by omitting transport-to-residential reallocation and weather-vulnerability margins. This distinction is central for designing energy-affordability and climate-adaptation policy in aging societies.

# A Supplementary Shift-Share IV Evidence

This appendix reports a supplementary survival-adjusted shift-share IV exercise. The baseline shift-share IV based on the full 65+ population share is limited by the open-ended 85+ age bin in Census age tables. The preferred survival-adjusted version therefore focuses on the 65–74 population share. This instrument starts from the year-2000 state age distribution and ages cohorts forward using national cohort progression in Census population estimates. The 65–74 version avoids the 85+ top-code problem and is interpreted as variation in entry into older age rather than the entire elderly stock.

The exclusion restriction is demanding. States with older baseline age structures may also differ in housing age, urban form, fuel infrastructure, climate adaptation, and retirement migration patterns. These factors can affect household energy outcomes independently of aging. State fixed effects absorb time-invariant differences and year fixed effects absorb common national shocks, but they do not fully address differential trends correlated with baseline age structure. The IV estimates are therefore interpreted as supplementary evidence about whether state-level predicted entry into older age points in the same direction as the household-level results.

Table 13: Survival-Adjusted Shift-Share IV Estimates

| Outcome | 2SLS Coef. on 65–74 Share | p-value | N |
|---|---|---|---|
| Log home energy | 5.0207 | 0.272 | 848 |
| | (4.5737) | | |
| Home energy expenditure | 8009.8382* | 0.067 | 848 |
| | (4368.4184) | | |
| Electricity share | 3.2827 | 0.170 | 848 |
| | (2.3911) | | |
| Fossil home-energy share | -3.2827 | 0.170 | 848 |
| | (2.3911) | | |
| High energy burden | -0.1867 | 0.862 | 848 |
| | (1.0759) | | |
| Energy burden, winsorized | -0.0946 | 0.890 | 848 |
| | (0.6817) | | |
| Gasoline share | -2.3271*** | 0.005 | 848 |
| | (0.8258) | | |

Notes: Standard errors are in parentheses. State-year 2SLS estimates instrument actual Census 65–74 population share with the survival-adjusted predicted 65–74 share based on year-2000 state age structure and national cohort progression. All specifications include state fixed effects, year fixed effects, average household size, average log income, owner share, and urban share. The first-stage coefficient is 0.240 (s.e. 0.055, p ¡ 0.001), with first-stage statistic 20.82. The estimand is interpreted as the effect of predicted entry into older age, not the full 65+ population stock. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

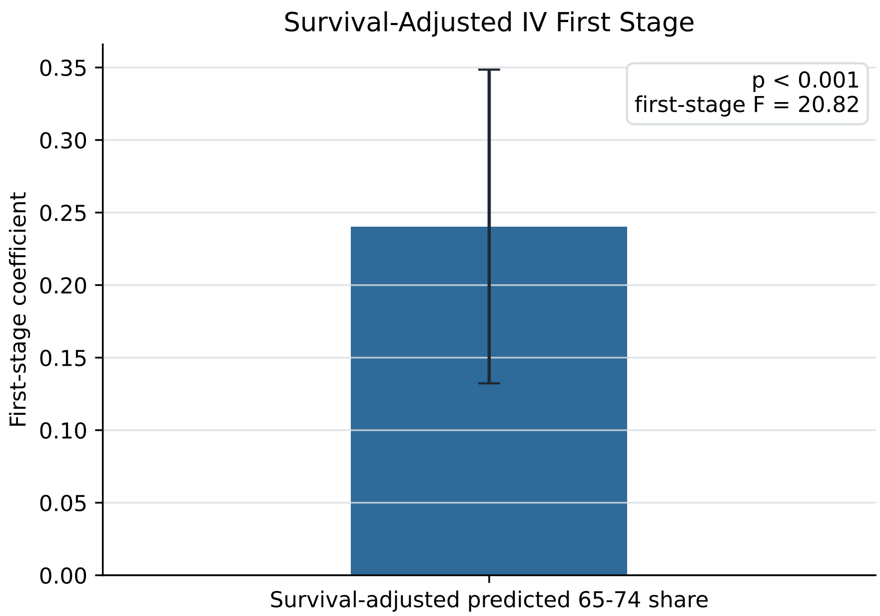


Figure 8: Survival-Adjusted Shift-Share IV First Stage

The IV results are consistent with the gasoline margin: predicted increases in the 65–74 share significantly reduce the gasoline share. Home-energy expenditure also rises at the 10 percent level. The IV estimates for residential fuel shares are not statistically significant, and the estimates for energy-burden outcomes are imprecise. Because of the exclusion-restriction concerns noted above, these estimates are not used as the paper's central causal evidence.

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